Bio-plausible Modeling of Associative Learning

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Abstract

Associative learning is key to the brain's learning process. Studying how this phenomenon takes place in the brain may shed light on many sophisticated functionalities in perception and cognition. Here, we propose a computational model of associative learning based on a brain-inspired structure and bio-plausible learning. We investigate two main associative learning procedures, namely classical conditioning, and operand conditioning. We discuss how our model forms these types of conditioning and what insights it can provide concerning behavioral experiments. We further examine the observed phenomena related to these conditioning experiments. Additionally, a new framework for spiking neural network simulation has been developed in this research that offers faster computations on GPUs.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(*Mahbod Nouri*)

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Chapter 1

Introduction

Associative learning happens when the brain identifies a connection between separate cues. A bacterium finding its way using a chemical gradient [38], bugs adjusting their foraging habits toward the regions with more available food, and a dog learning to follow instructions to receive a treat can all be examples of associative learning in living creatures. Understanding the process of associative learning in living creatures can shed light on more complex cognitive functions. This process is of high importance in cognition, perception, and learning. Associative learning encompasses many types of conditioning, such as classical conditioning (CC), operant conditioning (OC), and delay conditioning (DC).

In this study, we aim to investigate associative learning in a brain-like structure. The goal of this study is to find out how associative learning occurs in a biologically-plausible model of neural networks. To study this, we simulated a simplified model of natural neurons' dynamics along with experimentally observed biological learning rules. We tend to determine if a single neural model is able to demonstrate various conditioning behaviors as well as observed phenomena related to conditioning experiments.

To date, many computational models have been proposed to simulate this conditioning [33]. Nevertheless, a detailed study of the neural mechanisms of this phenomenon is still lacking. Most computational models that account for associative learning and conditioning are merely mathematical models that exclude any link to natural neural networks. Studies that have implemented conditioning on more bio-plausible architectures have also not demonstrated a detailed analysis of the learning process on the network. This study aims to use the recent progress in the field of computational neuroscience in order to construct a clear analysis of associative learning in a bio-plausible network and determine the effect of different involved parameters.

In the following sections, we first provide a brief history of previous works in the literature. Afterward, we provide information on the programming frameworks used for this study. We explain the setup of our classical conditioning and operant conditioning experiments and discuss the results of each study. Eventually, we conclude our findings and discuss how these insights can be beneficial in the fields of computational neuroscience, machine learning, and reinforcement learning.

Chapter 2

Background

2.1 Associative Learning

Several types of conditioning have been observed and suggested [7]. Here we focus on two main conditioning experiments, namely classical Conditioning and operant Conditioning.

2.1.1 Classical Conditioning

While examining dog digestion, Pavlov noticed a difference in the dog's bodily reactions to food. He found the dog was salivating in response to various noises (such as the food container) as long as the noises were consistent prior to the arrival of the meal. Pavlov set up an experiment in which he would ring a bell (conditioned stimuli - CS) shortly before feeding the dog (unconditioned stimuli - US) to explore this effect. Pavlov noticed that the dog salivated (unconditioned response - UR) to the bell sound after a while, even when there was no food present. This effect is now known as classical or Pavlovian conditioning [9].

2.1.2 Operant Conditioning

Animals' cognition and behaviors are formed around the consequences of their actions in their environment. For a living creature, it is of vital importance to be able to detect the process that leads to its benefit and survival advantage. Reward and punishment signals in the brain help the animal detect and repeat the action that leads to a gain or prevents a loss. These reward signals seem to be provided in the brain by neuromodulators, such as dopamine [13].

In an operant conditioning experiment, an animal would be rewarded (for example, given some food) or punished (for example, receiving an electrical shock) for performing certain actions. After a while, the animal learns to repeat the action that leads to the reward and avoid the behavior that leads to receiving punishment [38].

2.2 Modeling Associative Learning

Rescorla and Wagner [32] suggest a rule to model classical conditioning. The model is a variation of ”linear operator” models in which knowledge changes as a linear function of current knowledge. The Rescorla-Wagner model defines changes in association for each stimulus based on its own state and also the state of other stimuli. Van Hamme and Wasserman [17] modified the Rescorla-Wagner model in a way that the association between CS and US decreases if CS does not occur after the US. This can also model the effect of extinction. In these models, conditioning trials are assumed to be discrete events, so the temporal relationships between US and CS are ignored.

To counter this limitation, [34] used a simple neural network structure with one hidden layer to model the behavior in real-time instead of on a trial-to-trial basis. Temporal difference (TD) learning [40] extends the Rescorla-Wagner model to take into account the timing of the events. This approach helps an agent learn how to estimate a value for a given signal, which is based on future insight, hence it is widely used in reinforcement learning. Despite the fact that these models have been used in many cognitive studies, the fundamental differences between their structures and natural neural networks make a direct link between the two impossible.

2.3 Spiking Neural Networks

Spiking neural networks (SNNs) are designed based on a simplified architecture of biological neural networks' mechanisms. Each neuron in these networks is mimicking the functionality of a natural neuron to some extent. These neurons are connected through synapses, and they can interact with each other. When the membrane potential of a neuron reaches a certain threshold, a spike will be fired and the neuron will inject a specific amount of current into some post-synaptic neurons. Consequently, information can propagate through the network. Compared to artificial neural networks (ANN), spiking neural networks have been shown to demonstrate significantly less computational cost [27]. Low computational complexity and more similarity to brain

architecture have made SNNs interesting subjects to study for researchers in both the artificial intelligence (AI) and computational neuroscience fields.

2.3.1 Neuron model

Various models have been proposed to simulate neural dynamics, such as the Leaky Integrate-and-Fire (LIF) [14], Adaptive Exponential integrate-and-fire mode (AdEx), and the Hodgkin-Huxley [20] model. The main difference between these models is the level of complexity and how well they can resemble natural neurons. For example, the LIF model is one of the simplest neuron models with very low computational complexity, and it is therefore used in many studies that utilize SNNs for an AI application. Hodgkin-Huxley, on the other hand, involves a great level of biological detail and is able to predict a neuron's behavior with high accuracy. However, Hodgkin-Huxley neurons are computationally costly, and it is not feasible to stimulate a large network of these neurons. It would also be difficult to tune Hodgkin-Huxley neurons for AI purposes since their dynamics are determined by many variables.

In this research, we used the Izhikevich neuron model [21]. Izhikevich reduces the complexity of the Hodgkin-Huxley model to two differential equations. This neuron type can demonstrate diverse neuronal behaviors, such as bursting. Nevertheless, the computational complexity of the model is considerably lower, allowing it to carry out simulations with a large number of neurons. The dynamics of Izhikevich neurons are also controlled by merely four parameters.

2.3.2 Synaptic Plasticity

Konorski [1] first suggested that changes in synaptic strengths may be the underlying contributor to many cognitive functions such as learning and memory. Around the same time, in his groundbreaking research on functional causality, Donald Hebb argued that as a presynaptic neuron repeatedly stimulates a postsynaptic neuron to fire, it eventually becomes more efficient at driving the postsynaptic neuron [19]. This hypothesis is now known as the Hebbian rule for plasticity, and it greatly influenced neuroscience and machine learning in the following years. Other experimental research, such as that done by Bienenstock, Cooper, and Munro (BCM) [4] showed that if a presynaptic neuron just weakly drives a postsynaptic neuron, it can also lead to a depression of the synaptic strength.

By varying the relative frequency of the inputs to the neurons, Levi and Steward [25]

showed that the temporal order of inputs can be influential on synaptic modification. The research showed that when a weak input precedes a strong input by less than 20 ms, potentiation will occur in the synapse. Carrying out the same experiment in a reversed order, however, caused a depression in the synapse. After more experiments on the temporal relation of pre- and post-synaptic spikes [30] [28], the Spike-timing-dependent plasticity (STDP) rule was established [2]. Based on this rule, if a pre-synaptic spike comes shortly before a post-synaptic spike, a Long-Term Potentiation (LTP) occurs in the synapse. This is consistent with the Hebbian rule since a pre-post spike event can represent causality.Similarly, pre spike event causes Long-Term Depression (LTD) in the synapse. STDP is a biologically plausible learning rule applied to SNNs and its different variants are widely used in unsupervised and supervised learning [43]. Further studies have suggested how to modulate STDP learning to perform a goal-driven task [13].

2.4 Associative Learning in SNN

Izhikevich [22] used a reward-modulated STDP rule to demonstrate how a network of neurons is able to solve the distal reward problem in an operant learning task. The distal reward problem arises when a reward comes with a temporal gap and the network is supposed to assign the achieved reward to the current contributor. The author suggests an experiment in which several different neuron groups are activated randomly, and every time a target neuron group activates, a reward signal in the shape of an increase in DA level will be injected into the network with a 1-second delay. The paper shows that after several trials, the response of the network to the target neuron group considerably increases.

In [11], the authors used a group of three spiking neurons to demonstrate the classical conditioning effect: an action neuron, a US neuron, and a CS neuron. Also, by using a modulated version of STDP, they suggested a different network for operant learning. The study can show associative learning in small neural circuits. However, the structure and learning method used are not bio-plausible and cannot explain conditioning in more complex networks. In [31], researchers used a three-layer spiking neural network to model an insect olfactory pathway. The study uses reward and punishment signals so the network can associate olfactory cues with food and avoid distractor cues. To train the network, the study uses multi-spike tempotron(MST) [16] which is a gradient-based learning rule for spiking neural networks. Even though some studies have discussed

how a process such as backpropagation can take place in the brain [26], no explicit description or experimental data that can describe the gradient-based learning process in a biological neural network has been proposed.

Chapter 3

Methods

In this research, we construct a neural network based on biological inspiration. Here, we provide a detailed description of this network:

3.1 Neural dynamics

We use the Izhikevich neuron model for the simulations [21]. The Izhikevich neuron model follows two differential equations:

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I \tag{3.1}$$

$$\frac{du}{dt} = a(bv - u) \tag{3.2}$$

Where v and u are dimensionless variables that represent the membrane potential of the neuron and a membrane recovery variable, respectively, and t denotes time. a, b, c, and d are also dimensionless parameters that shape the dynamics of the neuron. When a neuron's membrane potential (v) surpasses a specific threshold ϑ, a spike will be emitted and the neuron's voltage and recovery will be updated to these values:

$$if\ v \geq \vartheta\ then\ v \rightarrow c\ \&\ u \rightarrow u + d \tag{3.3}$$

The term I indicates the injected current to each neuron and consists of two different terms:

$$I = I_{ext} + I_{syn} \tag{3.4}$$

Where I_{ext} is a current that determines the intrinsic firing rate of each neuron as well as any external stimulation, and I_{syn} denotes the synaptic current received from other

neurons in the network. The spontaneous activity of neurons is generated by a random input current with a mean μ_i and variance of σ_i.

Neurons interact with each other through synapses. When a neuron spikes, it leads to a chemical synaptic current that increases the voltage level in all post-synaptic neurons. The intensity of this increase is dependent on the strength of the synapse, which we call synaptic weight. After each spike, a value proportional to the synaptic weight will be added to the post-synaptic neurons' current, after a short time gap that accounts for the axonal delay [41]. For a post-synaptic neuron j, the input current from pre-synaptic neurons at each timepoint is equal to:

$$I_{syn} = \sum_i w_{ij} \delta(t - t_i^f - D) \tag{3.5}$$

Where w_{ij} denotes the synaptic weight from pre-synaptic neuron i to post-synaptic neuron j, D is the axonal delay, and t_i^f denotes the firing time for neuron i:

$$t_i^f = \{t | v_i \geq \vartheta\} \tag{3.6}$$

All simulations are carried out using the second-order Runge-Kutta method with a time-step $h = 1$ms. Excitatory neurons are of the regular-spiking type and constitute 80% of the neural population [21]. Inhibitory neurons are fast-spiking neurons and constitute the other 20% of the neurons. Parameters related to these neurons are described in Table 3.1.

3.2 Network

The proposed model is composed of N spiking neurons that can interact through synapses. Neurons are connected to each other based on an Erdos-Renyi (ER) random graph [12] where there is a p chance for the neuron i to have a synapse to the neuron j. We set $p = 10\%$, and there is no recurrent connections in the network, that is, $w_{ij} = 0$ for all $i = j$ at all time points.

3.3 Learning

In spiking neural networks, neural activity is shaped by external input and synaptic interactions between neurons. Learning takes place by adjusting the strength of these synapses. For our model, we use the STDP learning rule with an eligibility trace. Also, we use a reward-modulation scheme to train the network by providing rewards.

We only apply plasticity to excitatory neurons. The weights of all connections from inhibitory neurons are set to a fixed value, W_i, throughout the whole learning process.

3.3.1 Spike-timing-dependent plasticity (STDP)

We modify the strength of excitatory synapses using the STDP learning rule. Based on STDP, plasticity occurs based on the timing of the pre- and post-synaptic spikes. We apply STDP based on nearest neighbor spikes:

$$STDP(\Delta t) = \begin{cases} A_+ e^{-\frac{\Delta t}{\tau_+}} & : \Delta t \geq 0 \\ A_- e^{-\frac{\Delta t}{\tau_-}} & : \Delta t < 0 \end{cases} \tag{3.7}$$

Where $\Delta t = t_{post} - t_{pre}$ and t_{pre}, t_{post} represent time point in which pre- and post-synaptic neurons have spiked, respectively. A_+ and A_- are two parameters indicating maximum potentiation and depression for each spike pair, and τ_+ and τ_- are LTP and LTD time constants. Figure 3.1a shows this STDP window for a pair of spikes.

3.3.2 Eligibility trace

Upon STDP, we use eligibility trace to update weights. Each synapse in the network has an eligibility trace which is determined as follows:

$$\frac{de}{dt} = \frac{-e}{\tau_e} + STDP(\Delta t)\delta(t - t_{pre/post}) \tag{3.8}$$

Where $\delta(t)$ is the Dirac delta function, $t_{pre/post}$ denotes the time in which a pre- or post-synaptic spike has occurred, and τ_e is the eligibility time constant.

3.3.3 Reward Modulation

The learning process is modulated by the dopamine (DA) level in the network, which indicates a reward factor. We assume:

$$\frac{dM}{dt} = \frac{-(M - M_0)}{\tau_M} + M_R\delta(t - t_{rew}) \tag{3.9}$$

Where M is the dopamine level, M_0 is the resting value of dopamine (baseline dopamine level), τ_D is dopamine time constant, M_R indicates the intensity of the reward and finally, t_{rew} is the time that a reward signal is injected to the network. Synaptic weights will be changed based on the following equation:

$$\Delta w_{ij} = M\, e_{ij} \tag{3.10}$$

Parameter	Value
a_{ex}	0.2
b_{ex}, b_{in}	0.2
c_{ex}, c_{in}	-65
ϑ	30
D	10ms
μ_i	0
σ_i	3.5
A_+	1×10^-3
A_-	2×10^-3
τ_+	20ms
τ_-	20ms
τ_e	1s
τ_D	100ms
M_0	0.01
W_i	-1
γ	20

Table 3.1: Network hypermaeters used in the research. Subscripts *ex* and *in* denotes excitatory and inhibitory neurons respectively.

Figure 3.1b demonstrates these dynamics for a pair of neurons after a pre-post spike event.

3.3.4 Normalization

After every weight change, we constrain the total output weight from each excitatory neuron. For an excitatory neuron i, we multiply all output weights by a factor of $\gamma / \sum_j w_{ij}$, where γ is a scaling factor and represents the total output weight of neurons.

3.4 Hyperparameters

Most hyperparameters are chosen from previous studies and are designed to be in a bio-plausible range. The values of the hyperparameters used in this study are provided in Table 3.1

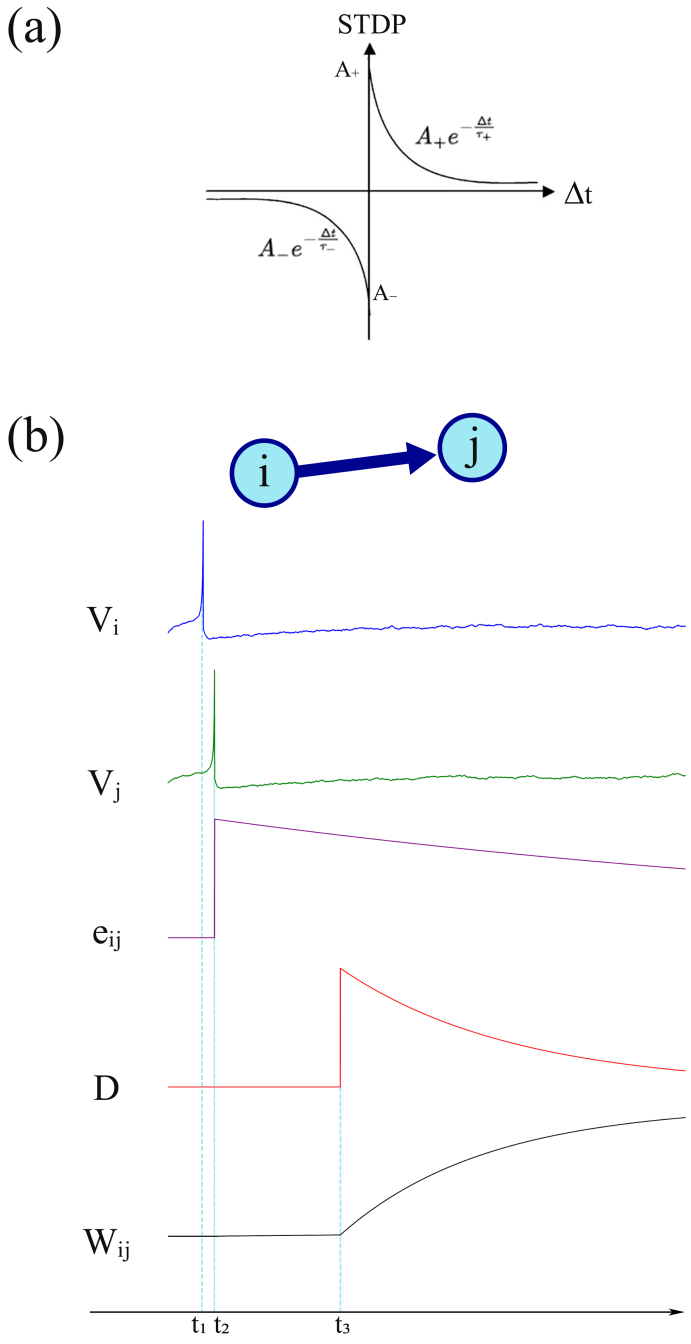


Figure 3.1: (a) An STDP window. After each pair of spikes, the synaptic eligibility trace will be updated based on this function. (b) Dynamics underlying synaptic plasticity of a connection from neuron i to neuron j. V_i and V_j show the voltage of neurons i and j respectively. Neuron i fires a spike at time t_1 and neuron j spikes at time t_2. Synaptic eligibility trace e_{ij} is updated at t_2 based on the STDP function in (a) with $\Delta t = t_2 - t_1$. A reward is injected into the network at time t_3. The synaptic weight W_{ij} will be increased proportionally to the eligibility trace of the synapse and the momentary reward in the network.

Chapter 4

Implementation

4.1 Coding Framework

Our research consists of simulations with long-term runs. Also, in order to study different hypotheses, we conducted many several simulations. In total, we simulated different models for more than 1,000,000 seconds (in simulation time). Therefore, the time complexity of the implementation has been of high importance. Studies on spiking neural networks are usually carried out using one of the established frameworks in Python, such as Brian 2[39] or BindsNET[18]. By using Pytorch [29] as the backend, BindsNET enables users to run their simulations on Graphic Processing Units (GPU) to accelerate the calculations. Based on a benchmarking study, BindsNET shows this approach leads to a considerable speed-up.

Inspired by BindsNET, we developed FastSpike framework. FastSpike also enables users to utilize the GPU for faster matrix multiplications. With some modifications, we were able to enhance the speed of our framework over BindsNET. This, however, comes at the price of more intensive memory usage. Therefore, when the load of the network gets close to the maximum limit of the memory, the framework's performance declines. Figure 4.1 shows a comparison between FastSpike and BindsNET for different set-ups. This benchmarking has been done on a Tesla P100 GPU. It can be seen that for the scope of networks that are used in this research, FastSpike can provide us with considerable acceleration over BindsNET.

FastSpike framework is publicly available at: https://github.com/mahbodnr/FastSpike

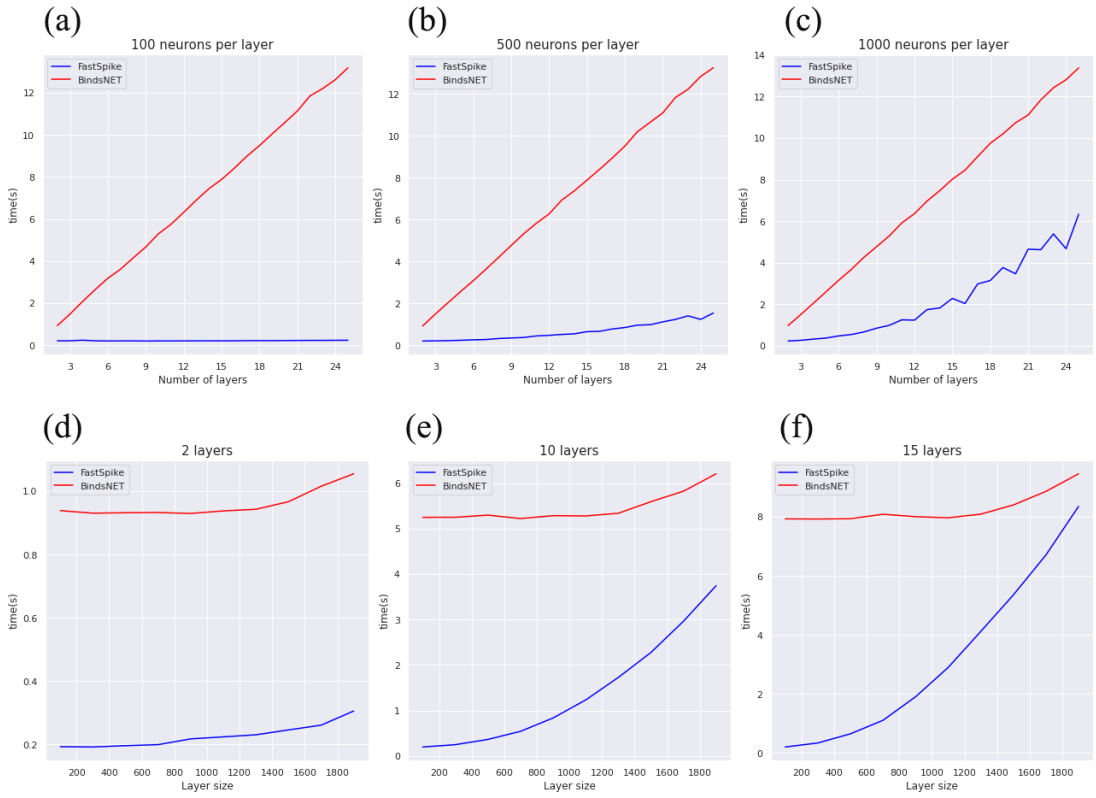


Figure 4.1: Benchmark results of FastSpike performance against BindsNET. Comparisons have been made based on similar fully-connected networks with random inputs. Upper row: varying the number of neurons in each neuron group in a network with a fixed number of neuron groups. Lower row: varying the number of layers with a fixed number of neurons per layer.

4.2 Julia Version

We have provided an implementation of the FastSpike framework, also in the Julia language. Julia combines a sophisticated LLVM compiler with a language design that focuses on performance [3], and it will also offer native GPU programming.

Besides the performance, we tried to make FastSpike easy to use. For example, the following sample code shows how to construct a network of leaky integrate-and-fire (LIF) excitatory and inhibitory neurons, connect them together, and simulate the network for 1000ms:

```
# set number on excitatory and inhibitory neurons
N_exc = 800
N_inh = 200
# construct a network with LIF neurons
net = Network(neurons = LIF(dt= 1))
# Add neuron groups
exc = add_group!(net, N_exc; name="Excitatory")
inh = add_group!(net, N_inh; name="Inhibitory")
# make a random connection with connection chance of 0.1
connection = randomConnection(N_exc, N_inh, 0.1)
# connections excitatory neurons to inhibitory neurons
connect!(net, exc, inh, connection)
# transform the network to the CUDA device
net = net |> gpu
# run the network with random input
for t in 1:1000
    run!(net; input_voltage = rand(1,N_exc+N_inh) |> gpu)
end
```

Until now, we have implemented LIF [14], Izhikevich [21], and adaptive exponential (AdEx) [5] neuron models. Also, concerning synaptic plasticity, we have implemented STDP, STDP with eligibility trace, Voltage-based STDP [8], and Calcium-based STDP [15] learning rules. The code is publicly available at: https://github.com/mahbodnr/FastSpike.jl

Chapter 5

Classical Conditioning

5.1 Conditioning Experiment

In a classical conditioning experiment, we investigate a stimulus that an animal is sensitive to as an Unconditioned Stimulus (US). This stimulus is associated with an Unconditioned Response (UR) in the animal. For example, in Pavlov's research, food was the unconditioned stimulus that could trigger salvation in the dog as a response. During the experiment, the presence of the US will be predicted by an irrelevant stimulus, for example, a bell ringing. This new stimulus is called Conditioned Stimulus (CS). Experiments revealed that after several trials, the animal shows the same response to the CS.

5.1.1 Experiment Setup

We examine this effect in our proposed model. We select three non-overlapping random groups of 25 neurons to represent CS, US, and UR. One can suppose that neurons in CS and US will be evoked by external stimuli while neurons in the UR group will trigger a physical or behavioral response. We manually set the weights from the US to UR to a large constant value. Accordingly, a burst of activity takes place in UR whenever the US gets stimulated.

During each trial of one second, first, the CS group will be simulated for 25ms. After a time gap of δ following the onset of CS stimulation, we stimulate the US neuron group also for 25ms. Figure 5.1 demonstrates a trial of this experiment. We use a network with $N = 300$ for this experiment. We set the M_0 to 1 and no rewarding process occurs during this experiment. Other excitatory neurons in the network are indicated

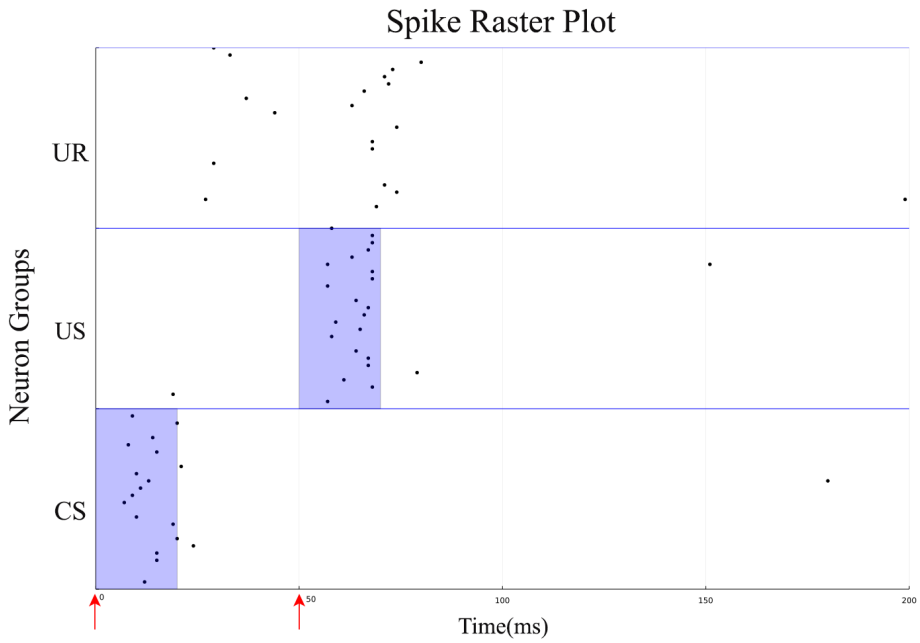


Figure 5.1: Classical conditioning experiment setup. Purple boxes indicate the presence of an external stimulus to the CS and US groups. Red arrows on the timeline indicate the onset of CS and US stimulus, respectively. We present the time interval between these two arrows with δ. This figure demonstrates an experiment with $\delta = 50$ms.

with S.

5.1.2 Results

Our results suggest that classical conditioning takes place for small time gaps between CS and US stimulation. Figure 5.3 shows the average output weights from the CS group throughout the training for $\delta = 0, 25, 50$ms. It is shown that CS connections toward US and UR neurons are potentiated through training, while connections toward other parts of the network are depressed. For $\delta = 0$ms CS and US neuron groups are stimulated simultaneously. Since UR is activated mainly through US neurons, these neurons will be activated slightly later than the onset of the stimuli, considering the axonal delay. As a result, weights from CS to UR will be more potentiated (since there is a higher probability of pre-post events for these connections). In other experiments, CS neurons are more strongly connected to US neurons.

To evaluate this effect in our model, we stimulate the CS subgroup (without stimulating the US) and investigate whether it can drive the UR neurons. Figure 5.2 shows group activations in a short time window after stimulating the CS, for different stages.

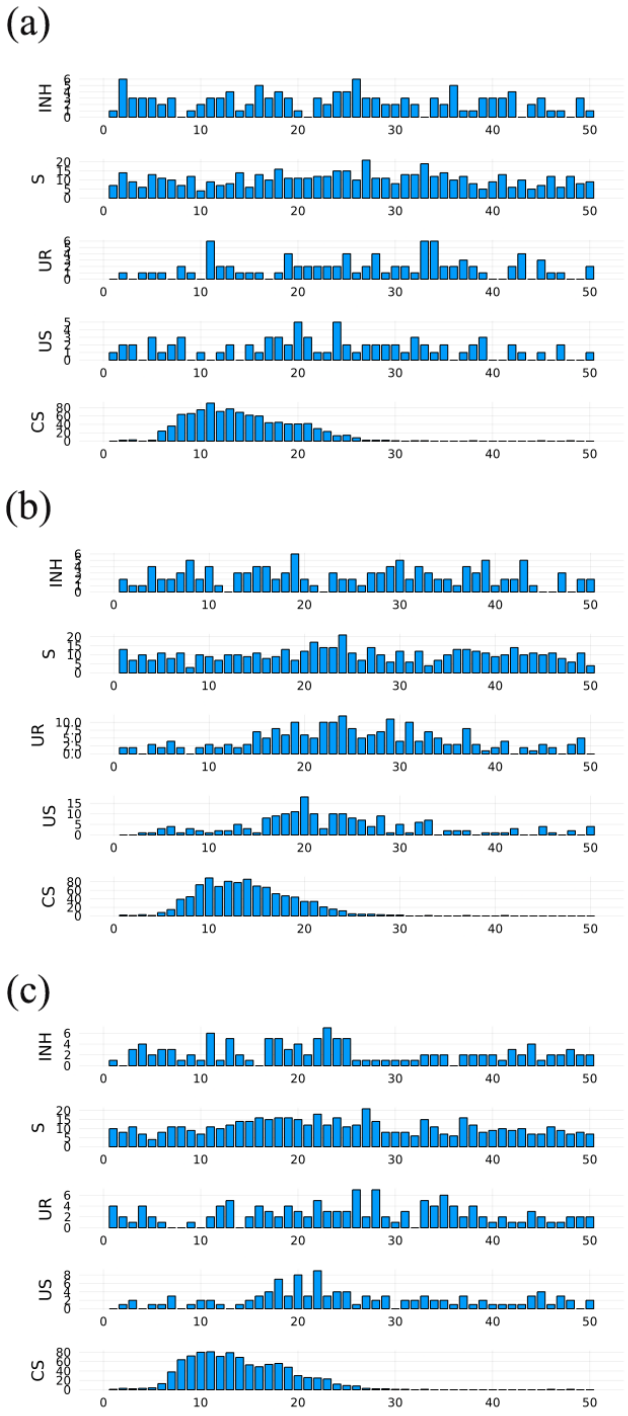


Figure 5.2: Spike response of neural groups in the network after only activating the CS neural group. Results were achieved after conducting 10 independent runs where we stimulated the CS neuron group in the first 25ms, and summed the number of spikes at each time point for the first 50ms. INH represents the inhibitory neurons in the network. (a) before the conditioning procedure, there is no significant change in US or UR spike pattern after the CS activation. (b) After classical conditioning, CS activity can trigger a considerable increase in both US and UR spikes, while not impacting the other neuron groups in the network. (c) After the extinction process, we again observed no significant correlation between the activation of CS and unconditioned neuron groups.

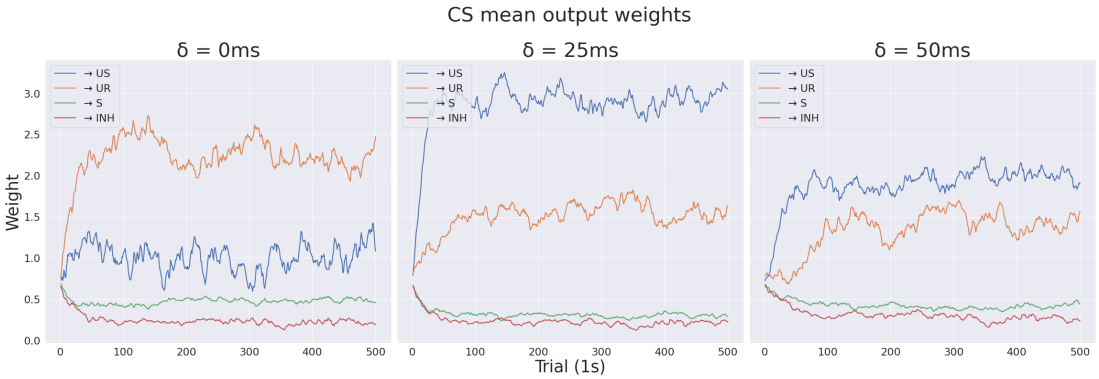


Figure 5.3: Average synaptic weights of connections from the CS neuron group to other subnetworks during the conditioning for different temporal gaps between stimuli. In all three figures, learning occurs before 100 trials, and the weights remain stable afterward.

We see that before the training, CS activation could not cause a considerable change in the UR activity level (Figure 5.2a). After 500 trials, the burst of activity in CS leads to a considerable increase in the number of spikes in the US and UR neuron groups (Figure 5.2b). This means the CS neurons can successfully predict the US and evoke the response in the network.

Despite the special case of $\delta = 0$ that discussed before, CS neurons are able to activate UR neurons by strengthening the connections with the US neuron group. This suggests that in classical conditioning, a CS does not trigger the response directly, but rather it will be associated with the US. Therefore, an unconditioned response happens because CS can evoke the US after conditioning, even in the absence of an external unconditioned stimulus. This observation may suggest that, for example, in Pavlov's experiment, the dog does not salvate to the bell ring directly, but rather the sound will evoke neurons that represent food in the dog's brain, and these neurons will then trigger salvation. Seeing the food itself can also follow the same argument since it can be considered as a visual stimulus that predicts the act of eating (which actually involves salvia), and thus it can activate the representation of food in the animal's brain.

5.1.3 Extending the Temporal Gap

Results show that when we increase the time gap between two stimuli, the conditioning effect will fade out. Figure 5.4a shows the average output weights of the CS neuron group during the training for $\delta = 100$ms. Also, Figure 5.4b shows the spike response to only stimulating CS neurons after this experiment. Both these results show that there is



Figure 5.4: The classical conditioning experiment with a 100ms temporal interval between stimuli. (a) The average synaptic weights of connections from the CS neuron group to other subnetworks follow a random and unstable trend throughout the conditioning process. (b) No considerable change occurs in US and UR neuron groups after the activation of the CS both before (left) and after (right) the training.



Figure 5.5: UR spikes after CS stimulation for experiments with different stimuli interval δ. To obtain the result, we ran each experiment for 100 trials. Afterward, we only stimulated the CS neuron group for 25ms and counted the number of UR spikes in the first 50ms. We conducted 10 separate runs and reported the average value of UR spikes.

no considerable change in the network after conducting the experiment representing the classical conditioning effect.

Figure 5.5 provides more details on the relation between conditioning and the temporal gap. Experiments also show that a large temporal interval between CS and US can disrupt conditioning, however, conditioning still occurs with intervals of orders of magnitude larger than the time gaps we demonstrated [10]. Considering the short STDP time constant that has been observed in experiments ($\approx$ 20ms [25]), for learning to happen with a large temporal interval (in the order of 100 seconds) between stimuli, the network requires a memory-like functionality. While this may become possible in networks with larger sizes, there is still much to be discovered about how memory takes place in neural networks.

One solution that was observed in experiments was to bridge the gap with oscillations. In some specific configurations, neuron groups would go through an oscillatory phase after the onset of the stimulus. The network would then learn the association through the interaction of neuron groups' activity waves (see appendix A). The problem with this solution was that it was dependent on the exact value of the temporal gap, and it would only work when the peaks of activity of CS were placed before the peaks of the US group. Therefore, conditioning would happen only for some specific time gaps, which is not consistent with experimental observations. Furthermore, this solution was

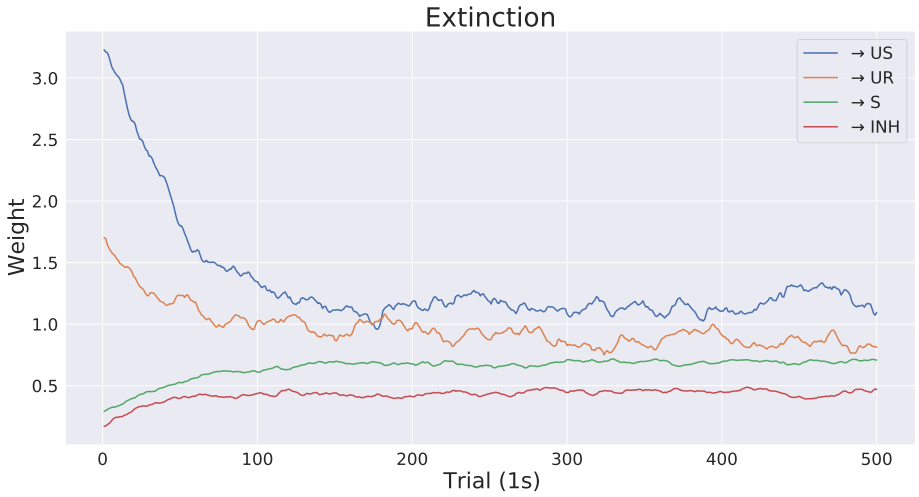


Figure 5.6: Average synaptic weights of connections from the CS neuron group to other subnetworks during an extinction procedure.

not reproduced in other configurations.

5.2 Extinction

In classical conditioning, conditioning gradually ceases when the CS is presented, but no US will arrive afterward, for several events [6]. In this case, since CS is no longer predicting the US, the association would gradually fade out and the animal would not show the response to the CS signal anymore. For example, Pavlov's dog eventually stopped salivating to the sound of the bell after they rang the bell but did not provide the food several times. This effect is called extinction.

Extinction was investigated in our proposed model. After the training phase, we simulate the network for several more trials, but this time we only stimulate the CS group at each trial. We observed that in all experiments, extinction occurs after multiple exposures of the network to the CS alone. Figure 5.6 shows the weight changes during an extinction experiment after the network was successfully conditioned. Furthermore, Figure 5.2c demonstrates the response of the network to stimulation of the CS neuron group after the extinction process. It is evident that the response of the network at this stage is similar to the network before the conditioning process (Figure 5.2a).

Chapter 6

Operant Conditioning

6.1 Response Learning

In an operant conditioning experiment, agents will be rewarded or punished for performing certain actions. Animals are able to detect a rewarding action by making an association between a stimulus and a reward signal. Here, we investigate the same effect in our bio-inspired model. We will select several random groups in the network, which represent different environmental states or different stimuli to the agent. The network will be rewarded if it can detect the correct corresponding response to each input.

6.1.1 Experiment Setup

To investigate this experiment in our model, we select n random non-overlapping groups of 50 neurons, called $S_1, S_2, ..., S_n$, representing different stimuli. We also select n other random groups of 10 neurons, called $R_1, R_2, ..., R_n$, as response groups. The objective of this experiment is for the network to learn to associate each stimulus group with the corresponding response group.

Every 1 second, we inject a short 2ms input pulse to a randomly selected stimulus group. We then select a 20ms time window, beginning from Dms after the pulse input, and count the number of spikes in all response groups. If the number of spikes in the response group that correspond to the selected stimulus group is more than all other groups, we count the trial as a successful one and inject a reward with the value of $M_0 = 2.0$ to the network. The reward will be injected after a short 10ms delay following the end of the window. This delay is meant to resemble the time gap that usually occurs between taking an action and receiving the reward in biological networks. For the

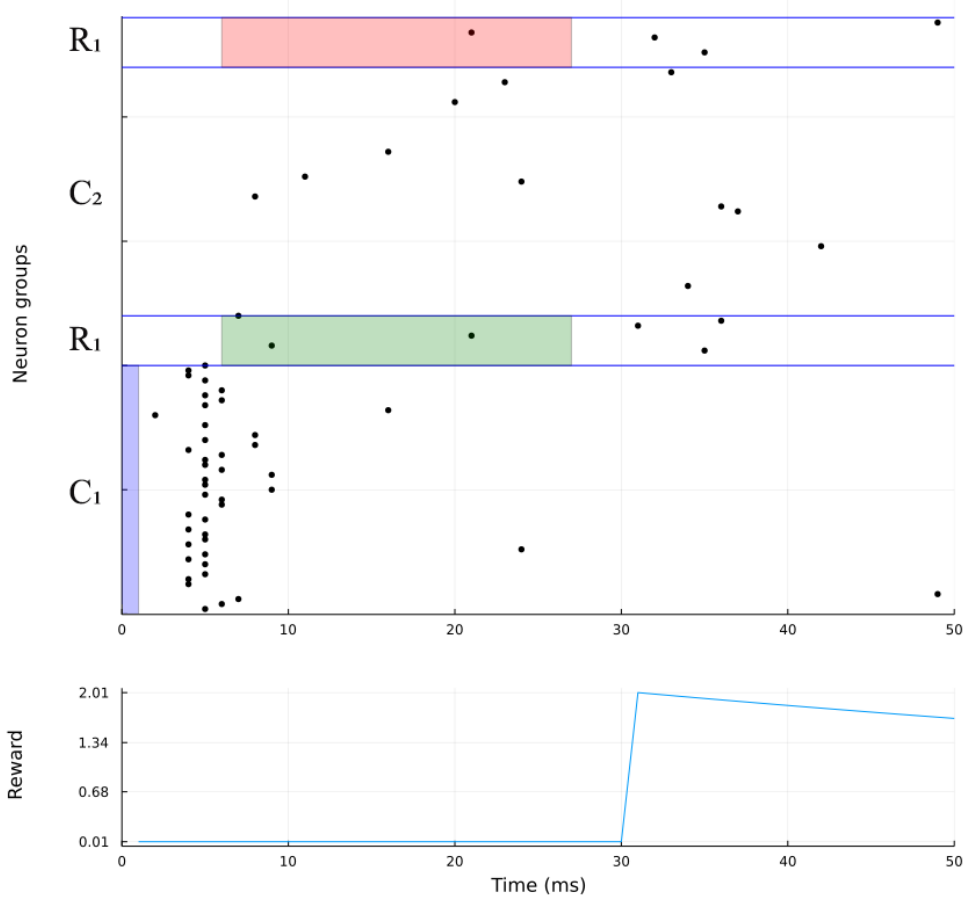


Figure 6.1: Operant conditioning experiment setup. The blue box indicates the $2ms$ pulse input to the randomly selected stimulus group (C_1 in this trial). We start counting response spikes Dms after this stimulus for 20ms which is indicated with green and red boxes on the figure. We count the number of spikes that are placed in these boxes and if the number of spikes in the correct response group (R_1 in this trial, which is shown by the color green) is greater than others, we consider the trial successful. In this case, a reward signal will be injected into the network after a 10ms gap following the counting period.



Figure 6.2: Model performance through the learning process in operant conditioning for experiments with different numbers of response groups.

sake of simplicity, we chose a small time gap. However, previous research shows that learning can properly progress even in the presence of much longer temporal delays [22]. If the number of spikes in the correct response group is less than or equal to any other group, we count the trial as an unsuccessful trial and no reward will be injected into the network. Figure 6.1 demonstrates a sample trial with this setup.

In this experiment, we consider each response group to represent a specific action of the agent. For example, all neurons in each group of R_i can be projected to a specific region in the motor cortex, triggering different sets of actions. In this case, the agent is learning to deliver the correct action to maximize its reward.

6.1.2 Results

To measure the functionality of the network through the learning process, we define a performance factor, $P(t)$, that for each trial t_n is equal to:

$$P(t_n) = \beta P(t_{n-1}) + (1-\beta)S \tag{6.1}$$

Where S is a binary value, indicating if the trial has been successful (1) or not (0), and β is a parameter that controls the significance of new results compared to the previous



Figure 6.3: Average synaptic weights of connections from each stimulus group to response groups during the training trials. We observe that each stimulus group C_m is more potentiating toward its corresponding response group R_m during the process.

ones. We set $\beta = 0.1$. By taking this approach, the performance factor indicates the model accuracy on the task in recent trials, which can better represent how well the network is performing the task than just considering the last trial. We have implemented the experiment on a network with $N = 1000$ for $n = 2,3,4$. Figure 6.2 shows the performance of these networks through training. This figure shows that in the first 1000 trials, the network gradually learns to activate the correct response group most of the time after receiving a stimulus for all experiments.

This mainly happens since the synaptic weights from the S_m group to the corresponding R_m group are getting potetiated durig the experiment ($1 \leq m \leq n$). Every time that S_m gets activated and then R_m shows a considerable rate of activity, the network gets a reward. This means many pre-post events that have happened in this trial from S_m to R_m will be potentiated with a much higher intensity. Figure 6.3 can demonstrate this by showing the average weight from each stimulus group S_m to the other response groups.

For a trial to be successful, it is also important for neurons in one stimulus group S_m, not to trigger neurons in other response groups. Therefore, synaptic weights from these neurons to other response groups must be depressed. This happens based on two main mechanisms. First, since the LTD rate is higher than the LTP rate in the network, the random activity of neurons leads to depression. As a result, since these connections usually do not lead to a reward for the network, the pattern of their activity remains mainly random and will be overall depressed. Another contributing mechanism here is weight normalization. Since each excitatory neuron has a limited total value for all of its outputs, potentiating toward a response group will automatically decrease the weights of connections with the other neurons.

To evaluate the robustness of our model, we also trained networks with $N = 300,500$. We didn't witness any considerable difference.

6.2 Adaptability

In order to survive in an ever-changing environment, living creatures must be able to adapt to changes in their environment. This includes updating their learnings with new data and changing their behavior when necessary. To evaluate this in our model, we designed another experiment. In this experiment, we have two stimulus groups, S_1 and S_2, that correspond to two response groups, R_1 and R_2, respectively. We train the network similar to the previous experiment, however, after the network is trained to

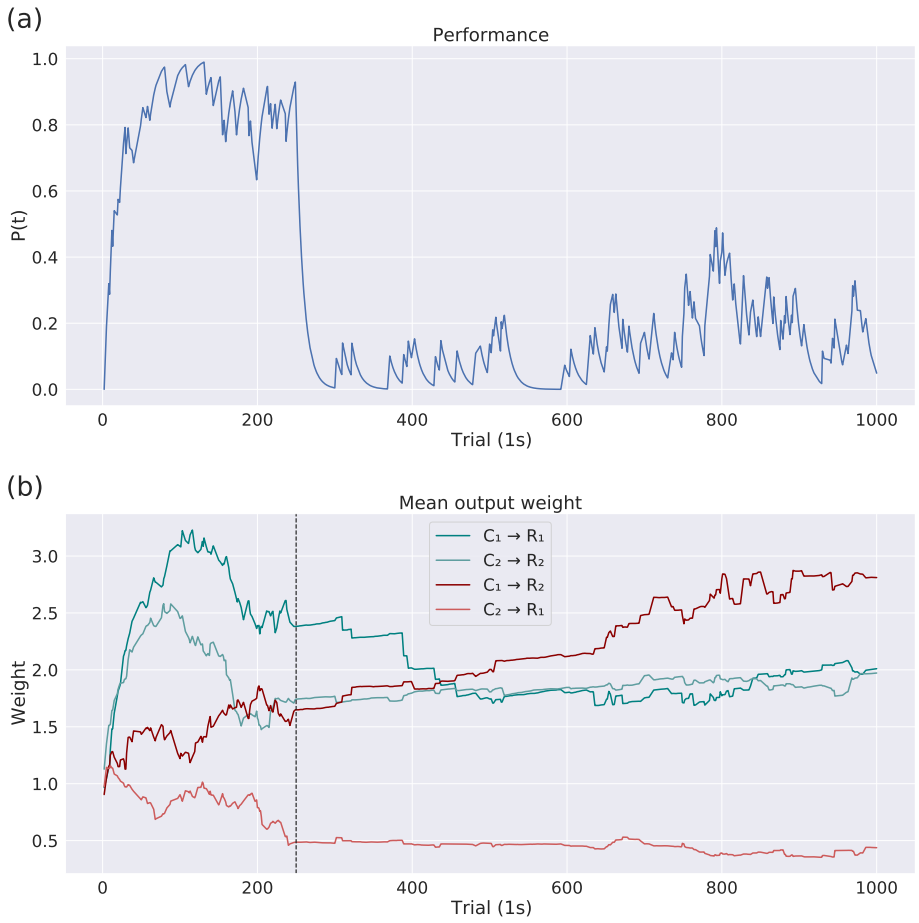


Figure 6.4: Switching response groups in an operant conditioning task with a normal rewarding method. We switched the correct corresponding group after 250 trials (dashed line). (a) The learning process after this trial becomes very slow and the model's performance cannot reach the previous level even after many more trials. (b) The rate of weight change becomes incredibly low after the switch trial due to the sparsity of rewards in the second phase. Average output weights show that the model is able to gradually associate C_1 to the new response group (R_2) after the switch. However, C_2 will remain more strongly connected to R_2 than R_1 in the second phase and this new association cannot be learned in the network.

select the correct response, we switch the response groups, that is, S_1 and S_2 must be able to activate R_1 and R_2 respectively to receive the reward.

Figure 6.4a shows the performance of the network during this experiment. It is evident from the results that the network is unable to adjust its performance after the change. This is because the network has already learned to activate R_1 and R_2 when stimuli S_1 and S_2 arrive with a high performance rate (¿90%). Therefore, when a switch occurs, events in which stimulus groups activate other response groups are scarce. Since the network is not receiving any reward until one of these scarce trials takes place, the new training process will be very slow and inefficient. Figure 6.4b suggests that however, we can see a learning process in one of the groups(C_1 to R_2 increases) this process is too slow. One approach to tackling this problem would be to use a higher value for the reward, to accelerate the learning in the second phase (after switching responses). A very high reward, however, can disrupt learning as changes become more intense, and any small noise can be influential.

6.2.1 Dynamic Rewarding

To address this issue, we adopted a dynamic reward-prediction error mechanism for rewarding. There is evidence to suggest that dopaminergic neurons in the brain release dopamine based on the difference between actual and expected rewards, rather than just on the actual reward [35]. Considering this, we define a dynamic value for the amount of dopamine that is rewarded to the network at each trial:

$$M_R(t_n) = \alpha(1 - P(t_n)) \tag{6.2}$$

Were M_R is the reward intensity in Equation 3.9, $P(t_n)$ is the performance factor at timepoint n, and α is a scalar factor, determining the maximum reward. Since the performance factor is a moving average that indicates how successful the network has been in the previous trials, it can be a good proxy for reward prediction in the current trial. The benefits of the dynamic rewarding method can be twofold. First, when the learning has not yet taken place (or there has been a change in environment) the reward-prediction is low, consequently, every successful trial leads to a high reward value and this can accelerate the learning. Secondly, when the training is conducted, the network performance increases and this means a higher reward-prediction, and thus, a lower amount of rewards in successful trials. This can improve the robustness by preventing the network from the influence of noise when the training has been successful.



Figure 6.5: Switching response groups in an operant conditioning task with a dynamic rewarding method. We switched the correct corresponding group after 250 trials (dashed line). (a) Despite the change in the environment, the learning still takes place at a considerable rate and the model is able to achieve its previous performance in the second phase (b) The rate of weight change is comparable with the first phase and the network can quickly learn the new configuration. In the second phase, both stimulus groups learn the new correct response groups (connections from C_1 to R_2 become stronger than R_1, and connections from C_2 to R_1 become stronger than R_2)

Figure 6.5a shows that dynamic rewarding can help the network to adjust to the changes in the environment. Training in the second phase takes place slower mainly because in the first phase weights start from a random initial value and average weights between each neuron group are close to each other (Figure 6.5b). On the other hand, in the second phase, connections from stimulus groups to old response groups are higher and must be depressed as well.

Chapter 7

Conclusion

In this research, we studied how associative learning can take place in a bio-plausible model. We proposed a method of spiking neural networks with synaptic plasticity that could mimic different learning tasks such as classical conditioning and operant conditioning. We showed that the network is able to associate a predictive signal with an unconditioned stimulus and its response. Using a reward signal, we trained the model to perform a desired action in response to a stimulus. Our model is capable of demonstrating more behaviors related to conditioning, such as extinction or adapting to changes.

Here, we first discuss how these findings can be useful in the realm of artificial intelligence. We then mention some caveats regarding our method and provide some suggestions on how to further this study.

7.1 Applications in Artificial Intelligence

Conditioning can also provide the opportunity for new methods of learning in AI. Currently, machine learning research mainly focuses on supervised learning, where a set of carefully labeled data must be provided to train a model. This labeling process is a time- and energy-consuming task. As well, a supervised model's performance will be limited to the specific type of labels that it has been fed with, meaning they usually fail to perform on a slightly different dataset after training.

On the contrary, associative learning can take place when a model simply receives information from its environment. The model will be able to detect patterns in the stimuli as it is able to associate representations of events that happen simultaneously, as shown in the classical conditioning experiment. This is closer to how humans learn from

the environment. A human baby, for example, learns to associate the feeling of thirst quenching with the visual representation of a glass of water after several spontaneous occurrences of these two events. Therefore, the baby will reach for a glass of water whenever he feels thirsty. Obviously, no explicit mathematical vector is required in order to learn this connection.

A bio-inspired AI model can take advantage of this fact, that is, a model that allows associative learning can gain knowledge based on mere exposure to information in an environment. As an example, in a visual task, instead of providing tons of labeled images, a model can be trained by being fed random videos. The model will gradually learn to make meaningful associations between its inputs. For instance, the visual representation of an apple may be gradually associated with the sound of the word "apple", since there is a high chance that these two stimuli will occur close to each other. Furthermore, the learning will not be restricted to the specific types of labels anymore, and the network can gain a wide range of learning (for instance, a concept of motion in the real world) from the data.

How to use this collected data, however, will be challenging in such models. One way could be to use a reward signal for the model in order to achieve a goal, as demonstrated in the operant conditioning experiment. This way, the model learns to gain useful knowledge from the environment to perform the particular task and get the reward. This model can therefore be used as a reinforcement learning (RL) agent. Its advantage over other RL methods can be that, unlike these methods [44], this model is not over-relied on the reward signal to be trained and it would be able to extract useful information in an environment even in the absence of a dense reward signal.

7.2 Limitations

We proposed a network to account for both classical and operant conditioning experiments. There are some limitations to our approach. Most importantly, the experiments are successful only when the time gaps between different phases are short enough to be supported by the STDP time window. In the Classical Conditioning experiment, as discussed in Section 5, learning will be disrupted when the temporal interval between stimuli reaches merely 100ms. Similarly, for the Operant Conditioning experiment, we only considered short periods, such as pulse input and a short time window to account for the response. In contrast, experiments conducted on animals show that conditioning occurs with much larger temporal intervals [6].

In this research, we assumed that the representation of an external stimulus can be represented as a burst of activity in a neural group. For example, in the Classical Conditioning experiment, we stimulated a subgroup of neurons to represent conditioned or unconditioned stimuli. Although this is a common approach in similar studies, there is no solid evidence that representations of stimuli in the brain are encoded in this way alone. In this study [23], for instance, the author suggests a time-locked (but not synchronous) activity of subgroups as a representation of different concepts in the network. As a result, the temporal resolution of spikes becomes more important in this approach.

Another limitation of our study was the computational complexity of the model. We investigated networks of up to 1000 neurons, however, this is orders of magnitude smaller compared to an animal nervous system [24]. It is possible that larger neural networks will display more complex behaviors that may enhance or disrupt the conditioning we observed in this study. Considering the quadratic relation between the size of the network and its computational complexity, the simulation of a large neural network was out of the scope of this study. Nevertheless, we can expand the scope of the study by utilizing neuromorphic hardware, that can provide a fast and energy-efficient platform for large-scale spiking neural networks [42].

7.3 Future Works

In the previous section, we mentioned a few improvements that could be made to extend this study, such as using larger networks. Here, we discuss a few more approaches that we believe have the potential for continuing this research.

We performed the two main conditioning experiments, namely classical conditioning, and operant conditioning. To date, several more conditioning experiments have been suggested, such as delay conditioning and trace conditioning [7]. Investigating these procedures can determine the robustness and generality of the proposed model. It is worth noting that implementing these experiments can be particularly challenging since the network must learn the correct timing to perform the task. This may require developing a chain of activities in order to reach a target group at the right time.

During the experiment, we used a simple and established variant of STDP. Nevertheless, studies show that this learning rule cannot fully describe the experimental data on synaptic plasticity [36]. To address this issue, several plasticity models with higher levels of bio-plausibility have been suggested, including Voltage-based STDP [8], and

Calcium-based STDP [15]. It is possible for more complicated learning rules to result in more complex network behaviors. Future studies can investigate if these learning methods can help resolve any of the current limitations.

Lastly, the model can be implemented to solve more complex problems. For example, the model can control an agent in an environment through the level of activity of neuron groups. Conditioning tasks can then be investigated on this agent. In this way, a more realistic scenario can be used to evaluate the robustness of the proposed model.

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Appendix A

First appendix

A.1 Oscillatory Behavior

Here we demonstrate a specific network configuration that leads to oscillatory behavior in subnetworks after the onset of stimuli. These oscillations can help the network bridge the temporal gap between two stimuli, therefor learning with larger stimulus intervals can occur in the network. However, conditioning only happens while using certain time intervals. This is due to the fact that these oscillations form waves of activity in the network. If the phase of these waves in the CS is behind the US and UR, their interaction would result in more pre-post events for connections from the CS to these neuron groups. On the other hand, when the CS activity wave phases the US and UR waves, there is a higher chance of pos-pre events, and connections from CS to the US and UR will be depressed.

For this experiment, we use an external baseline input with $\mu_i = 4$ and $\sigma_i = 1$. This can lead the network toward more oscillatory behaviors. Instead of the normalization technique, we used an additive STDP learning rule [37] in this experiment. Figure A.1 illustrates spike raster plots and activity levels of neuron groups in one trial at the end of the training process for two experiments with $\delta = 125$ and $delta = 175$. We can see that in the first experiment, oscillations in CS are further than US and UR in phase. On the other hand, in the second experiment, the peaks of activity in CS were placed behind the peaks of activity in the US and UR due to the larger time interval. Results in the Figure A.2 confirm this observation. As we see in this figure, the average output weight for connections from CS neurons to US and UR will be depressed in the first experiment and potentiated in the second one.

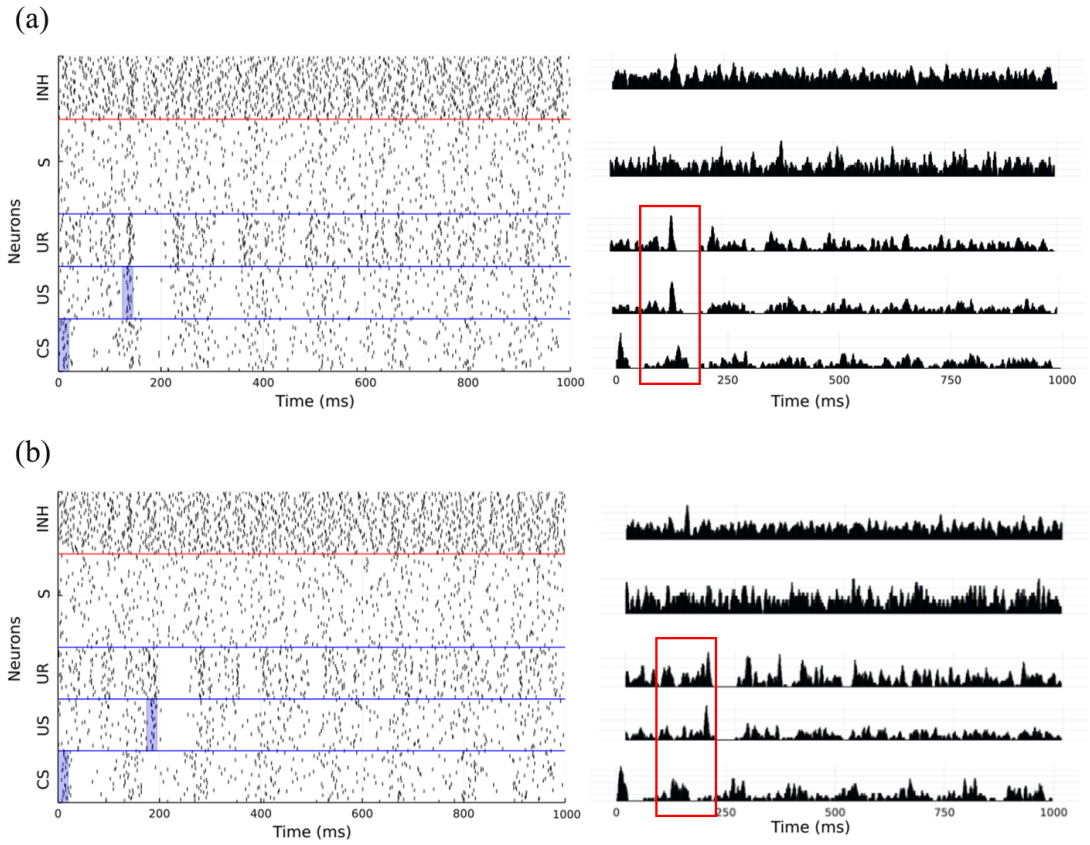


Figure A.1: Spikes raster and activity level of neuron groups after classical conditioning for experiments with (a) $\delta = 125$, and (b) $\delta = 175$. Red boxes indicate the first interaction of CS oscillation with the oscillation wave in the US and UR groups. It shows that in (a), CS reaches the peak of activity after the US and UR, while this is the opposite in (b).

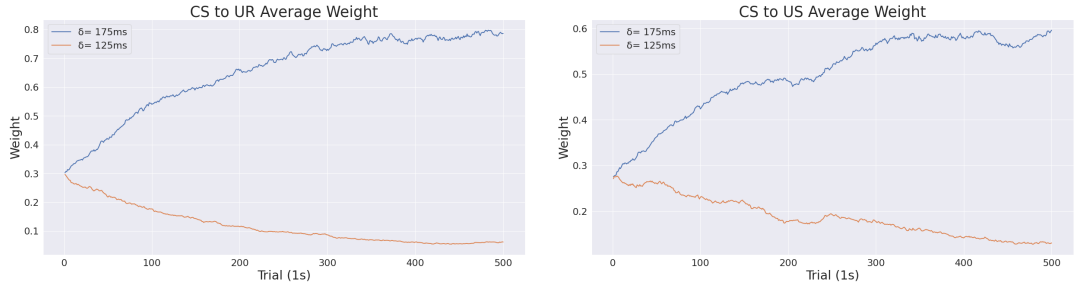


Figure A.2: Average synaptic weights of connections from CS neurons to (a) UR and (b) US neuron groups for two experiments with different temporal gaps between stimuli δ.